Causal Inference with Ordinal Outcomes: Flexible Estimation Using Normalizing Flows

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Latent Traits and Ordinal Measurement

- Political science wants to evaluate causal effects on continuous, latent traits Y_i^* (e.g., support for policies, trust in institutions)
- However, we routinely observe observed, ordinal survey indicators $Y_i \in \{1, \dots, J\}$ (e.g., a Likert scale).

Notation and Variables:

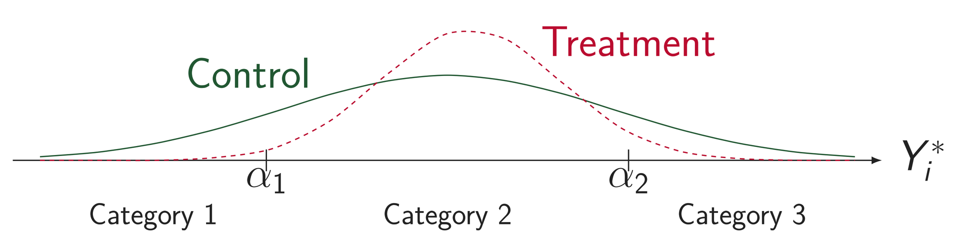
- $D_i \in \{0, 1\}$: Binary treatment indicator.
- X_i: Vector of baseline covariates.
- $m(X_i, D_i)$: Systematic latent index model (under a linear model, $m(X_i, D_i) = \tau D_i + X_i^\top \beta$, where τ represents the latent treatment effect).
- ϵ_i: Latent error term with CDF $F(\cdot)$.
- α: Vector of thresholds in a latent space.

We can write the model in latent space as:

$$Y_i^* = m(X_i, D_i) + \epsilon_i$$

We observe Y_i via a threshold-crossing mechanism partitioning the latent scale:

$$Y_i = j \iff \alpha_{j-1} < Y_i^* \le \alpha_j, \qquad j = 1, \dots, J$$



Why Cardinalization Fails (Toy Example: The Sign Flips)

A common approach is to assign arbitrary numerical scores $c(j)$ to the categories. The cardinalized treatment effect is then the difference in expected values:

$$\tau_c = \mathbb{E}[c(Y_i(1)) - c(Y_i(0))] = \sum_{j=1}^{J} c(j) \left[\Pr(Y_i(1) = j) - \Pr(Y_i(0) = j)\right].$$

Suppose treatment redistributes the probability distribution across three ordered categories from Control to Treatment as follows:

$$\Pr(Y_i(0) = j) = (0.30,\ 0.40,\ 0.30) \quad \longrightarrow \quad \Pr(Y_i(1) = j) = (0.15,\ 0.65,\ 0.20).$$

Then two strictly monotone codings starting from 1 yield opposite signs:

$$c_A = (1, 2, 3): \quad \tau_{c_A} = 1 \cdot (-0.15) + 2 \cdot 0.25 + 3 \cdot (-0.10) = +\mathbf{0.05},$$
$$c_B = (1, 2, 6): \quad \tau_{c_B} = 1 \cdot (-0.15) + 2 \cdot 0.25 + 6 \cdot (-0.10) = -\mathbf{0.25}.$$

Failure of Latent Scale Identification

- Because the location and scale of Y_i^* are unobserved, we can apply any positive affine transformation that yields same observed ordinal distribution ($\tilde{Y}_i^* = a + cY_i^*$ and $\tilde{\alpha}_j = a + c\alpha_j$ with $c > 0$):

$$\Pr(\tilde{\alpha}_{j-1} < \tilde{Y}_i^* \le \tilde{\alpha}_j) = \Pr(a + c\alpha_{j-1} < a + cY_i^* \le a + c\alpha_j) = \Pr(\alpha_{j-1} < Y_i^* \le \alpha_j).$$

- The latent index component is scaled by c ($\tilde{m} = cm$).
- The latent thresholds and covariate coefficients are shifted and scaled accordingly.

Theorem 1

The latent systematic index $m(X_i, D_i)$ is identified only up to an arbitrary scale shift $c > 0$ and location shift a.

Distributional Effects (What We CAN Identify)

Focus instead on what is identified: observed-scale probability distributions.

Target Causal Estimands:

$$\text{Category-specific effect: } \Delta_j = \Pr(Y_i(1) = j) - \Pr(Y_i(0) = j)$$

Theorem 2

Under standard conditions (SUTVA, Conditional Ignorability $Y_i(d) \perp D_i \mid X_i$, and Positivity), the potential outcome distribution is fully identified:

$$\Pr(Y_i(d) = j) = \mathbb{E}_X\left[\Pr(Y_i = j \mid D_i = d, X_i)\right]$$

Proposition 1

Based on Theorem 2, any causal quantity that is a functional of this distribution including category-specific effect is also identified.

Standard Estimation: Parametric Ordered Probit/Logit

If the latent error distribution $F(\cdot)$ of ϵ_i is known to be Standard Normal or Standard Logistic, estimating the conditional probabilities is straightforward.

- **Ordered Probit:** Assumes $F(\cdot) = \Phi(\cdot)$ (standard normal CDF).
- **Ordered Logit:** Assumes $F(\cdot) = \Lambda(\cdot)$ (standard logistic CDF).

We maximize the sample log-likelihood to estimate standard parameters $\Theta = \{\Theta_m, \alpha\}$:

$$\ell_n(\Theta) = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{J} \mathbf{1}(Y_i = j) \log\left[F(\alpha_j - m(X_i, D_i)) - F(\alpha_{j-1} - m(X_i, D_i))\right]$$

The Core Challenge: Unknown Error Distribution

- **Unknown Base Distribution:** In real-world applications, the true latent error distribution $F(\cdot)$ is **never known** in advance.
- **Complex Latent Shapes:** Latent attitudes in political science can be polarized or skewed, directly violating standard symmetric and unimodal parametric shapes.

Forcing standard normal or logistic errors when the true latent shape is complex leads to biased and unreliable causal effects Δ_j.

The Core Idea: Normalizing Flows

How can we estimate the shape of $F(\cdot)$ directly from the data?

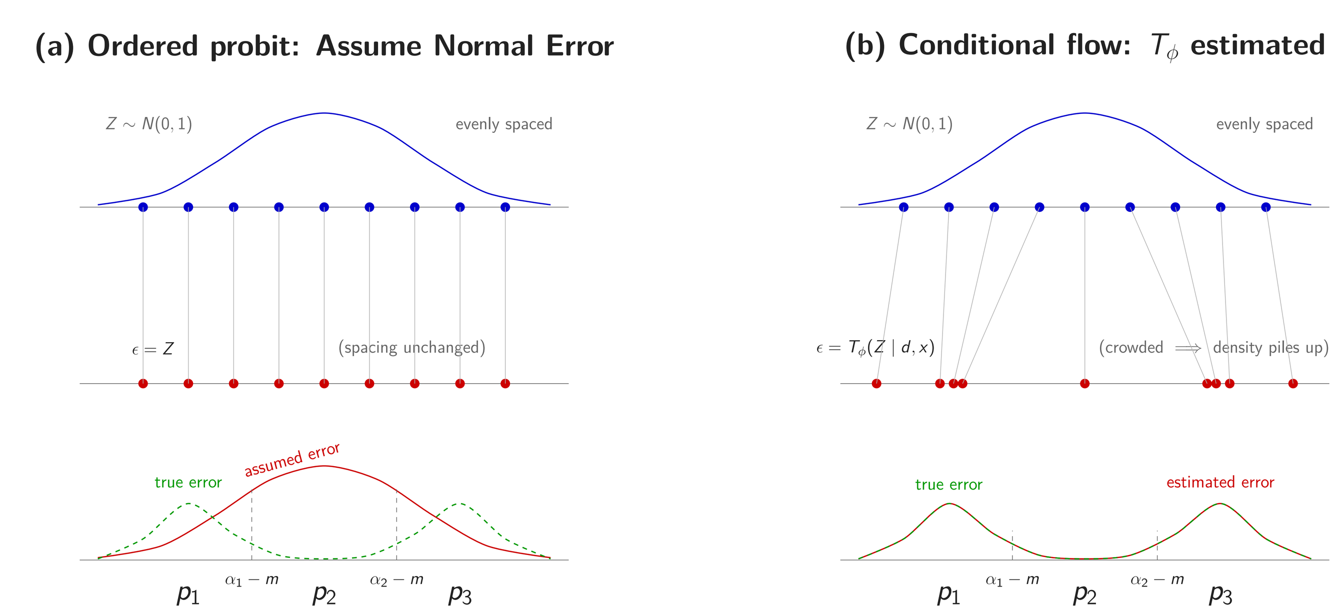
- A **Normalizing Flow** is an invertible map T_ϕ that stretch and warp the base distribution to match the true error distribution (ϵ_i):

$$\epsilon_i = T_\phi(Z_i \mid D_i, X_i)$$

  where ϕ is a vector of parameters for the flow model.
- This gives us the cumulative distribution function (CDF) of the true error distribution analytically:

$$F(\epsilon_i \mid d, x) = \Phi\left(T_\phi^{-1}(\epsilon_i \mid d, x)\right)$$

- Among flow models, monotone rational-quadratic splines (Durkan et al., 2019) retain invertibility and permit analytical CDF evaluation.



Likelihood and Consistency

Substituting the flow-based CDF, the parameters $\Theta = \{\Theta_m, \phi, \alpha\}$ maximize the sample log-likelihood for $m = m(X_i, D_i)$:

$$\ell_n(\Theta) = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{J} \mathbf{1}(Y_i = j) \log\left[\Phi\left(T_\phi^{-1}(\alpha_j - m \mid D_i, X_i)\right) - \Phi\left(T_\phi^{-1}(\alpha_{j-1} - m \mid D_i, X_i)\right)\right]$$

Assumption C.3 (Flow Approximation & Identification): The family of distributions that can be approximated by the flow is rich enough to approximate the true error CDF arbitrarily well.

Assumption C.4 (Regularity Conditions): The parameter space is compact, the likelihood satisfies standard continuity and differentiability conditions.

Theorem C.2 (Consistency of the Flow Estimator)

Suppose the latent outcome model and threshold structure are correctly specified, and Assumptions C.3–C.4 hold. Then, as $n \to \infty$, the maximum likelihood estimator is consistent up to a positive affine transformation:

$$\|\hat{\Theta} - \Theta_0\| = o_p(1).$$

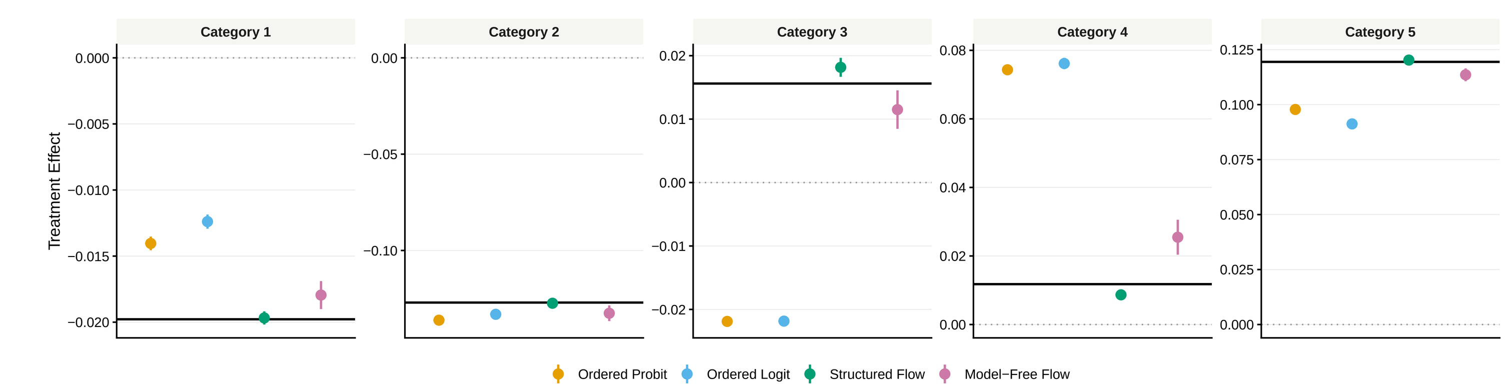
- Therefore, the flow-based estimators consistently estimate Δ_j

Nonlinear Latent Model Extension

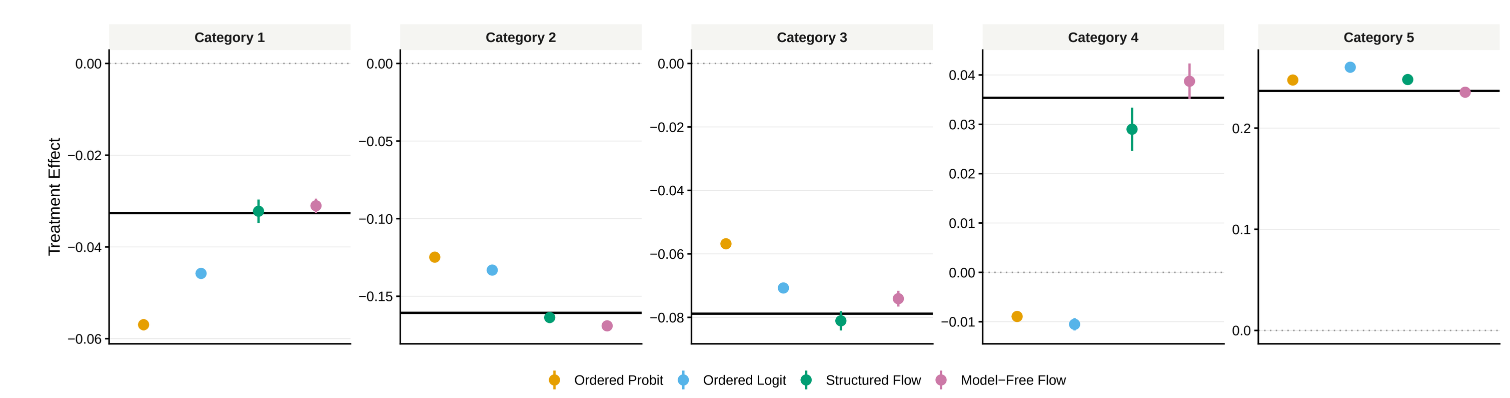
If $m(X_i, D_i)$ is nonlinear, the model-free normalizing flow T_ϕ can condition directly on a nonparametric representation of (D_i, X_i) without altering the likelihood setup.

Monte Carlo Simulation

- **Polarization:** a bimodal latent error distribution.
- **Nonlinearity:** nonlinear treatment-response structure.



Category-specific effects under a polarized latent distribution ($n = 2000$).



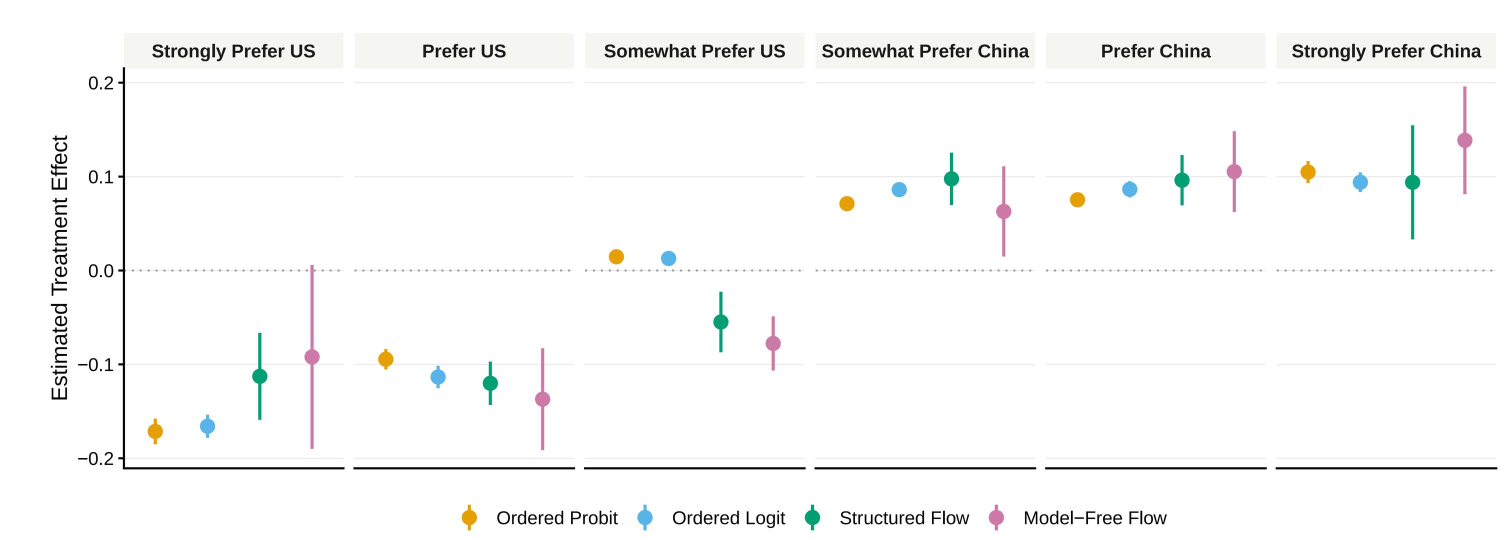
Category-specific effects under a nonlinear structure ($n = 2000$).

- Parametric error misspecification distorts category-level effects.
- Structured flows improve robustness with little efficiency loss.
- Model-free flows are most flexible but noisier in finite samples.

Application: Authoritarian Propaganda

Replicating Mattingly et al.(2025)'s survey experiment on how Chinese state media affects public opinion on relative preference for the United States versus China.

- **Parametric Models (Probit/Logit)** overestimate the conversion of citizens into extreme supporters (*Strongly Prefer China*).
- **Flexible Flow Estimators** show that the propaganda's primary impact is concentrated in **intermediate categories**.
- **Implication:** Authoritarian propaganda is persuasive not by creating extreme ideologues, but by softening opposition and winning over moderate/indifferent citizens.



Conclusion

1. **Cardinalized mean effects are not identified with ordinal outcomes.**
2. **Category-specific effects are identified under standard identification assumptions.**

Report distributional effects and compare structured and model-free estimators when ordered logit/probit assumptions are doubtful.

Contact and Materials

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Code & Data: `https://github.com/chanhyuk58/ordinal_flow`